UNIVERSITY OF SRI JAYEWARDENEPURA - FACULTY OF APPLIED SCIENCES

BSc Degree First Year First Semester Course Unit Examination – Sept/Nov 2025

DEPARTMENT OF COMPUTER SCIENCE

ICT 105 1.5 Computer Architecture

Time: One and a half (1 ½) hours **No. of questions: 03** **No. of pages: 02** **Total marks: 100**

Answer ALL questions

Question 01 [30 Marks]

i. Modern processors often use a modified Harvard architecture instead of a pure Von Neumann or Harvard architecture.

a. Briefly explain what a modified Harvard architecture is, and how it differs from pure Von Neumann and Harvard models. [6 marks]

b. Discuss two key advantages of using a modified Harvard architecture in modern processors. [4 marks]

c. Explain how the use of caches in a modified Harvard architecture helps improve instruction throughput and overall CPU performance. [5 marks]

ii. Explain the functions of the Arithmetic Logic Unit (ALU) in detail. Include at least four operations it can perform. [8 marks]

iii. Compare assembly language and high-level language in terms of readability, portability, and execution. [7 marks]

Question 02 [35 Marks]

i. a. Illustrate the instruction execution cycle of a computer with the help of a diagram. [5 marks]

b. Briefly explain the role of the PC, IR, MAR, and MDR. [8 marks]

ii. Consider the instruction ADD R1, R2 where the initial values of R1 = 50 and R2 = 200. Describe step-by-step how this instruction is executed. [6 marks]

iii. Explain the difference between single-bus architecture and multi-bus architecture. [6 marks]

iv. Modern computers use a memory hierarchy instead of a single large memory.

a. Draw and label a typical memory hierarchy. [6 marks]

b. Explain how temporal locality and spatial locality are exploited in cache memory. [4 marks]

Question 03 [35 Marks]

i. A calculator displays the results in both binary and decimal. Consider the 6-bit signed binary numbers:

$A=001101_2$, $B=000111_2$

Showing all intermediate steps, perform the subtraction A−B using the signed-2's complement method.
[6 marks]

ii. Consider the following instruction:

LOAD R4, 50(R3)

a. Identify the addressing mode used. [4 marks]

b. If R3 = 1000, show the step-by-step calculation of the effective memory address accessed.
[5 marks]

iii. MIPS32 follows a load-store architecture. Explain why this design choice simplifies pipelining.
[5 marks]

iv. Consider the following MIPS32 instructions:

```
add $t1, $s1, $s2
lw $t2, 100($s3)
j 40000
```

a. For each instruction, identify its type format (R-type, I-type, J-type). [3 marks]

b. Show the instruction encoding fields as appropriate. [6 marks]

c. Translate the MIPS32 assembly instructions into MIPS32 machine instructions. [6 marks]

Supplementary details

Name	Register Number
$zero	0
$v0-$v1	2-3
$a0-$a3	4-7
$t0-$t7	8-15
$s0-$s7	16-23
$t8-$t9	24-25

Name	Register Number
R-type	000000
lw	100011
j- type	000010
sw	101011
add	100000

****